QUESTION ANALYSIS

EEE475

RF and Microwave Engineering

Question Analysis

Sessions 2020–21 to 2018–19 *3 papers, 77 questions, 138 formulas*

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1. At a glance:

This analysis covers **3 papers** (2020-21, 2019-20, 2018-19), with 77 questions and 452 marks offered. It shows what past papers asked. It does not predict the next paper, and with so few papers a pattern can easily be chance.

A short-notes question is split across its notes, each taking an equal share of the marks, so some marks are not whole numbers.

The five topics with the most marks:

Topic	Marks	Share	Questions	Top subtopics (marks)
Antennas	145	32 %	23	Hertzian dipole (25); Linear arrays (16); Radiation from accelerating charges (13)
Transmission lines	86	19 %	13	Terminated lines (30); Line equations (17); Line constants (11)
Impedance matching	43	10 %	7	Single-stub matching (28); Quarter-wave transformer (15)
Microwave systems	42	9 %	8	Radar (13); Radio link budget (11); Receivers and mixers (6)
Smith chart	28	6 %	4	Constant-r circles (18); Circle properties (6); Reading impedance (4)

In the topic pages, a formula name in bold is a key formula, the number after a subtopic is its marks offered, and a blue symbol in an equation links to its meaning in the symbol list, where the arrow leads back to the equation.

2. Heat map:

Marks offered, by topic and paper, in book order. A darker cell means more marks. Click a topic to open its page.

Topic and subtopic	2020-21	2019-20	2018-19	All
EM waves	**6**	·	·	**6**
Plane waves	**6**	·	·	6
Transmission lines	**31**	**34**	**21**	**86**
Terminated lines	**12**	**4**	**14**	30
Line equations	**8**	**9**	·	17
Line constants	**11**	·	·	11
Lossy lines	·	**10**	·	10
Open and short tests	·	·	**7**	7
Slotted-line measurement	·	**7**	·	7
Reflection coefficient	·	**2**	·	2
Standing wave ratio	·	**2**	·	2
Smith chart	**12**	**12**	**4**	**28**
Constant-r circles	**12**	**6**	·	18
Circle properties	·	**6**	·	6
Reading impedance	·	·	**4**	4
Waveguides and microstrip	**12**	**9**	**6**	**27**
Microstrip	**5**	·	**6**	11
Rectangular waveguide modes	**7**	·	·	7
TM waves	·	**6**	·	6
Circular waveguide modes	·	**3**	·	3
Network analysis	·	**11**	·	**11**
S-parameters	·	**6**	·	6
Signal flow graphs	·	**5**	·	5
Impedance matching	**19**	**10**	**14**	**43**

Topic and subtopic	2020-21	2019-20	2018-19	All
Single-stub matching	**14**	**6**	**8**	28
Quarter-wave transformer	**5**	**4**	**6**	15
Resonators	**3**	**2**	·	**5**
Cavity resonators	**3**	**2**	·	5
RF components and filters	·	**12**	**12**	**24**
RF behaviour of L and C	·	**6**	**6**	12
Two-port parameters	·	·	**6**	6
Filters	·	**6**	·	6
Amplifier design	·	**14**	**12**	**26**
Constant-gain circles	·	**9**	·	9
Oscillators	·	·	**7**	7
Stability	·	·	**5**	5
Constant-gain condition	·	**5**	·	5
Microwave tubes and devices	**9**	·	·	**9**
Klystron	**2.25**	·	·	2.25
Magnetron	**2.25**	·	·	2.25
Gunn diode	**2.25**	·	·	2.25
Travelling wave tube	**2.25**	·	·	2.25
Microwave systems	**16**	**9**	**17**	**42**
Radar	·	**7**	**6**	13
Radio link budget	·	·	**11**	11
Receivers and mixers	**6**	·	·	6
AM power	**5**	·	·	5
EIRP	**5**	·	·	5
RF hazards	·	**2**	·	2
Antennas	**42**	**39**	**64**	**145**
Hertzian dipole	**25**	·	·	25
Linear arrays	·	**7**	**9**	16
Radiation from accelerating charges	**13**	·	·	13
Antenna aperture	·	**6**	**6**	12
Near and far field	·	**7**	**5**	12
Pattern calculations	·	**5**	**6**	11
Remote sensing	·	**5**	**5**	10
Directivity and beam area	·	**9**	·	9
Magnetic dipole	·	·	**8**	8
Horn antennas	·	·	**7**	7
Paraboloid reflectors	·	·	**6**	6
Effective height and area	**4**	·	·	4
Reciprocity	·	·	**3**	3
Image theory	·	·	**3**	3
Polarization	·	·	**3**	3
Yagi-Uda antenna	·	·	**3**	3

3. Priority list:

Must know: Antennas (145), Transmission lines (86), Impedance matching (43), Microwave systems (42), Smith chart (28), Waveguides and microstrip (27)

Likely: Amplifier design (26), RF components and filters (24), Resonators (5)

Less often: Network analysis (11), Microwave tubes and devices (9), EM waves (6)

The rule: Must know means asked in every paper. Likely means asked in at least half the papers. Less often means asked in fewer than half. Numbers in brackets are marks offered, and each list runs from most marks to fewest.

4. Topic pages:

4.1. EM waves:

6 marks, 1 question, in 1 of 3 papers. Book: Pozar Ch. 1; Sadiku Ch. 10.

Most often numerical (1 of 1). Subtopics: Plane waves.

Formulas:

Subtopic	Formula	Equation	Used in
Plane waves (6)	**Wave speed**	$u = \dfrac{c}{\sqrt{\mu_r \epsilon_r}}$	20-21 3(a)
	Intrinsic impedance	$\eta = \eta_0 \sqrt{\dfrac{\mu_r}{\epsilon_r}}$	20-21 3(a)
	Wavelength	$\lambda = \dfrac{2\pi}{\beta}$	20-21 3(a)

Symbols:

μ_r, ϵ_r = relative permeability and permittivity ↩ η = ratio of E to H (Ω) ↩ λ = wavelength (m) ↩

Questions:

No.	Paper	Marks	Type	Question
Plane waves				
1	20-21 3(a)	6	numerical	A plane wave in a nonmagnetic medium has $\vec{E} = 50\sin(10^8 t + 2z)\hat{a}_y$ V/m. Find: (i) The direction of wave propagation (ii) λ, f, and ϵ_r (iii) $\vec{H}$

4.2. Transmission lines:

86 marks, 13 questions, in 3 of 3 papers. Book: Pozar Ch. 2; Sadiku Ch. 11.

Most often numerical (6 of 13). Subtopics: Terminated lines; Open and short tests; Lossy lines; Reflection coefficient; Standing wave ratio; Line equations; Slotted-line measurement; Line constants.

Formulas:

Subtopic	Formula	Equation	Used in
Terminated lines (30)	**Reflection coefficient**	$\Gamma_L = \dfrac{Z_L - Z_0}{Z_L + Z_0}$	20-21 1(c), 19-20 6(b), 18-19 3(c)
	Standing wave ratio	$S = \dfrac{1 + \lvert\Gamma\rvert}{1 - \lvert\Gamma\rvert}$	20-21 1(c), 19-20 4(a), 18-19 3(c)
	Voltage maximum	$d_{max} = \dfrac{\theta\lambda}{4\pi}$	18-19 3(c)
	Input impedance	$Z_{in} = Z_0 \dfrac{Z_L + jZ_0 \tan\beta l}{Z_0 + jZ_L \tan\beta l}$	18-19 3(c)

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Subtopic	Formula	Equation	Used in
	Short-circuited line	$V = -2jV_0^+ \sin\beta z,\ I = \dfrac{2V_0^+}{Z_0}\cos\beta z$	18-19 4(c)
	Short and open input	$Z_{sc} = jZ_0\tan\beta l,\ Z_{oc} = -jZ_0\cot\beta l$	19-20 6(c), 18-19 4(c), 18-19 5(d)
	Load from Γ	$Z_L = Z_0\dfrac{1+\Gamma}{1-\Gamma}$	20-21 1(c)
	Input impedance	$Z_{in}(l) = \left(\dfrac{Z_L - jZ_0\tan\beta l}{Z_0 - jZ_L\tan\beta l}\right)Z_0$	20-21 1(c)
Open and short tests (7)	Z_0 from two measurements	$Z_0 = \sqrt{Z_{oc}Z_{sc}}$	19-20 6(c), 18-19 5(d)
	βl from two measurements	$\tan^2\beta l = -Z_{sc}/Z_{oc}$	18-19 5(d)
Lossy lines (10)	Lossy input impedance	$Z_{in} = Z_0\dfrac{Z_L + Z_0\tanh\gamma l}{Z_0 + Z_L\tanh\gamma l}$	19-20 2(d)
	γ from a short	$\gamma l = \dfrac{1}{2}\ln\dfrac{1+T}{1-T},\ T = \tanh\gamma l = \dfrac{Z_{sc}}{Z_0}$	19-20 2(d)
Reflection coefficient (2)	Along the line	$\Gamma(z) = \Gamma_L e^{2\gamma z},\ \Gamma(d) = \Gamma_L e^{-j2\beta d}$	19-20 4(a)
	Return loss	$RL = -20\log_{10}\lvert\Gamma\rvert$ dB	19-20 4(a)
Line equations (17)	Telegrapher's equations	$\dfrac{dV}{dz} = -(R+j\omega L)\,I,\ \dfrac{dI}{dz} = -(G+j\omega C)\,V$	20-21 1(a), 19-20 6(b)
	General solution	$V(z) = V_0^+e^{-\gamma z} + V_0^-e^{\gamma z}$	20-21 1(a), 19-20 6(b)
	Line current	$I(z) = \dfrac{1}{Z_0}\left(V_0^+e^{-\gamma z} - V_0^-e^{\gamma z}\right)$	19-20 6(b)
	Propagation constant	$\gamma = \alpha + j\beta = \sqrt{(R+j\omega L)(G+j\omega C)}$	20-21 1(a), 19-20 6(b)
	Characteristic impedance	$Z_0 = \dfrac{R+j\omega L}{\gamma} = \sqrt{\dfrac{R+j\omega L}{G+j\omega C}}$	20-21 3(b), 19-20 6(b)
	Low-loss line	$\alpha \approx \dfrac{R}{2Z_0} + \dfrac{GZ_0}{2},\ \beta \approx \omega\sqrt{LC},\ Z_0 \approx \sqrt{\dfrac{L}{C}}$	20-21 1(a)
Slotted-line measurement (7)	Phase from the minimum	$\theta_\Gamma = 2\beta d_{min} - \pi$	19-20 6(d)
	Impedance at extrema	$Z_{min} = \dfrac{Z_0}{S},\ Z_{max} = Z_0S$	19-20 6(d)
	Load from SWR	$Z_L = Z_0\dfrac{1 - jS\tan\beta d_{min}}{S - j\tan\beta d_{min}}$	19-20 6(d)
Line constants (11)	Conductor loss	$\alpha_c = \dfrac{R}{2Z_0},\ \alpha_d = \alpha - \alpha_c$	20-21 1(b)
	Shunt conductance	$G = \dfrac{2(\alpha - \alpha_c)}{Z_0}$	20-21 1(b)
	Phase velocity	$u_p = \dfrac{\omega}{\beta}$	20-21 3(b)

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Symbols:

Γ_L = reflection coefficient at the load ↩

Z_L = load impedance (Ω) ↩

Z_0 = line impedance (Ω); characteristic impedance (Ω) ↩

Γ = reflection coefficient at the load ↩

S = SWR, $1 \leq S < \infty$ ↩

θ = angle of Γ_L (rad) ↩

d_{max} = distance from the load ↩

l = line length (m); distance from the load (m) ↩

β = phase constant (rad/m) ↩

V_0^+ = incident amplitude (V)

Z_{oc}, Z_{sc} = measured input impedances (Ω) ↩

d = distance from the load (m), lossless ↩

$|\Gamma|^2$ = fraction of power reflected

R, L, G, C = line constants per metre (Ω/m, H/m, S/m, F/m)

R = series resistance (Ω/m) ↩

L = series inductance (H/m) ↩

G = shunt conductance (S/m) ↩

C = shunt capacitance (F/m) ↩

V_0^+, V_0^- = incident and reflected amplitudes (V); forward and backward amplitudes (V) ↩

α = attenuation constant (Np/m) ↩

d_{min} = distance of the first voltage minimum (m) ↩

α_c, α_d = conductor and dielectric parts of α (Np/m) ↩

u_p = phase velocity (m/s) ↩

Questions:

No.	Paper	Marks	Type	Question
Terminated lines				
1	20-21 1(c)	12	derivation	In Fig. 1, a lossless transmission line is shown where, $V(z) = V_0^+(e^{-j\beta z} + \Gamma e^{j\beta z})$, $I(z) = \frac{V_0^+}{Z_0}(e^{-j\beta z} - \Gamma e^{j\beta z})$, Γ is the reflection coefficient at point $z = 0$, Z_0 is the characteristics impdedance of the line, Z_L is the load impedance, and β is the phase constant. Show that (i) $\Gamma = \frac{Z_L - Z_0}{Z_L + Z_0}$ (ii) $Z_{in}(l) = \left(\frac{Z_L - jZ_0 \tan \beta l}{Z_0 - jZ_L \tan \beta l}\right) Z_0$
2	19-20 6(c)	4	theory	Briefly describe input impedance for open circuit termination and short circuit termination.
3	18-19 3(c)	7	numerical	A lossless transmission line length 0.434λ and characteristic impedance 100 Ω is terminated in an impedance $260 + j180$ Ω. Find the voltage reflection coefficient, standing wave ratio and the location of maximum voltage on the line.
4	18-19 4(c)	7	derivation	Derive the equation for a short circuit terminated transmission line and sketch voltage, current and impedance as a function of line length.
Open and short tests				
5	18-19 5(d)	7	numerical	A lossless transmission line of length 1.5 m, which is less than quarter wavelength has open circuit impedance $-j54.6$ Ω and short circuit impedance $j103$ Ω. Find Z_o and γ. Also find the input impedance if the line length is doubled and the length of line if it is to appear as an open circuit at the input.
Lossy lines				
6	19-20 2(d)	10	numerical	The input impedance of a short-circuited lossy transmission line of length 2 m and characteristics impedance 75 Ω is $15 + j120$ Ω. Find α and β of the line. Determine the input impedance if the sort-circuit *[sic: should be short-circuit]* is replaced by a load impedance $Z_L = 37 + j45$ Ω.
Reflection coefficient				
7	19-20 4(a)i	2	theory	Write short notes on the following topic (i) Reflection coefficient
Standing wave ratio				
8	19-20 4(a)ii	2	theory	Write short notes on the following topic (ii) Standing wave ratio
Line equations				
9	20-21 1(a)	8	derivation	From the lumped-element equivalent circuit, derive the expression of the propagation constant in terms of series resistance R, series inductance L, shunt conductance G, and shunt capacitance C per unit length. Deduce this expression to low-loss transmission line.
10	19-20 6(b)	9	derivation	From the general equation of transmission line find the expressions for characteristic impedance and reflection coefficient.

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No.	Paper	Marks	Type	Question
Slotted-line measurement				
11	19-20 6(d)	7	numerical	The standing wave ratio on a lossless 50 Ω transmission line terminated in an unknown load impedance is found to be 2. The distance between the successive voltage minima is 15 cm, and the first minimum is located at 5 cm from the load. Determine reflection coefficient, load impedance and equivalent length and terminating resistance of a line such that the input impedance is equal to Z_L.
Line constants				
12	20-21 1(b)	5	numerical	If the propagation constant of a transmission line is $50\angle 89.38°$ m^{-1}, $R = 4$ Ω m^{-1}, $L = 0.5$ μH m^{-1}, $C = 200$ pF m^{-1}, calculate the value of shunt conductance with proper unit.
13	20-21 3(b)	6	numerical	A telephone line has $R = 30$ Ω km^{-1}, $L = 100$ mH km^{-1}, $G = 0$, and $C = 20$ μF km^{-1}. At $f = 1$ kHz, obtain: (i) The characteristic impedance of the line (ii) The propagation constant (iii) The phase velocity

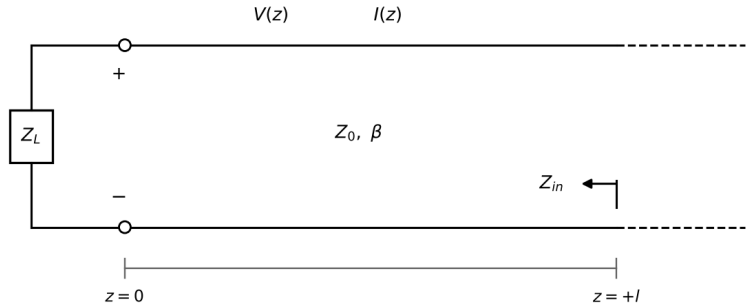


Figure 1: 20-21 1(c): Fig. 1 of the paper (redrawn)

4.3. Smith chart:

28 marks, 4 questions, in 3 of 3 papers. Book: Ludwig and Bretchko Ch. 3.

Most often theory (2 of 4). Subtopics: Reading impedance; Constant-r circles; Circle properties.

Formulas:

Subtopic	Formula	Equation	Used in
Reading impedance (4)	**Input impedance**	$Z_{in} = Z_0 \dfrac{Z_L + jZ_0 \tan \beta l}{Z_0 + jZ_L \tan \beta l}$	18-19 3(b)
Constant-r circles (18)	Impedance and Γ	$z = r + jx = \dfrac{1+\Gamma}{1-\Gamma}$	19-20 2(a)
	Constant-r circle	$\left(u - \dfrac{r}{1+r}\right)^2 + v^2 = \left(\dfrac{1}{1+r}\right)^2$	20-21 2(a), 19-20 2(a)
	Normalized impedance	$z = \dfrac{Z}{Z_0} = r + jx,\ \Gamma = \dfrac{z-1}{z+1}$	20-21 2(a)
Circle properties (6)	**Constant-x circle**	$(u-1)^2 + \left(v - \dfrac{1}{x}\right)^2 = \left(\dfrac{1}{x}\right)^2$	20-21 2(a), 19-20 2(b)

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Symbols:

l = line length (m) ↩

β = phase constant (rad/m) ↩

r, x = normalized resistance and reactance ↩

u, v = real and imaginary parts of Γ ↩

Questions:

No.	Paper	Marks	Type	Question
Reading impedance				
1	18-19 3(b)	4	theory	Write the steps to determine line input impedance using Smith Chart.
Constant-r circles				
2	20-21 2(a)	12	derivation repeats 19-20 2(a)	Derive the equation of the constant resistance and constant reactance circle of a Smith Chart. You should start form the relation between the reflection coefficient and the normalized load.
3	19-20 2(a)	6	derivation	Show that the normalized resistances represent circles in reflection coefficient plane.
Circle properties				
4	19-20 2(b)	6	theory	Write the properties of r-circles and x-circles of a Smith chart.

4.4. Waveguides and microstrip:

27 marks, 5 questions, in 3 of 3 papers. Book: Pozar Ch. 3; Sadiku Ch. 12.

Most often theory (3 of 5). Subtopics: Microstrip; Circular waveguide modes; TM waves; Rectangular waveguide modes.

Formulas:

Subtopic	Formula	Equation	Used in
Circular waveguide modes (3)	TE wave impedance	$Z_{TE} = \dfrac{k\eta}{\beta}$	19-20 2(c)
	Circular guide cutoff	$f_c = \dfrac{p\,c}{2\pi a\sqrt{\epsilon_r}}$	19-20 2(c)
TM waves (6)	TM field	$E_z = E_0 \sin\dfrac{m\pi x}{a} \sin\dfrac{n\pi y}{b}\, e^{-j\beta z}$	19-20 5(a)
	Cutoff wave number	$k_c = \sqrt{\left(\dfrac{m\pi}{a}\right)^2 + \left(\dfrac{n\pi}{b}\right)^2},\ k_c^2 = k^2 - \beta^2$	19-20 5(a)
	Phase constant	$\beta = \sqrt{k^2 - k_c^2} = k\sqrt{1 - \left(\dfrac{f_c}{f}\right)^2}$	19-20 5(a)
	Cutoff frequency	$f_{c,mn} = \dfrac{1}{2\sqrt{\mu\epsilon}}\sqrt{\left(\dfrac{m}{a}\right)^2 + \left(\dfrac{n}{b}\right)^2}$	20-21 3(d), 19-20 5(a)
	TM wave impedance	$Z_{TM} = \dfrac{E_x}{H_y} = \dfrac{\beta}{\omega\epsilon} = \eta\sqrt{1 - \left(\dfrac{f_c}{f}\right)^2}$	19-20 5(a)

Symbols:

a = radius (m)

a, b = guide width and height (m) ↩

$k = \omega\sqrt{\mu\epsilon}$ = wave number (rad/m)

m, n = mode indices ↩

Questions:

No.	Paper	Marks	Type	Question
Microstrip				
1	20-21 6(a)	5	theory repeats 18-19 4(b)	What are the uses of a microstrip line? Mention its advantages. Describe the type of losses in a microstrip line.
2	18-19 4(b)	6	theory	What is microstrip antenna? Mention some of the applications of microstrip line.
Circular waveguide modes				
3	19-20 2(c)	3	theory	Describe how TE mode is different from TM mode in a circular waveguide.
TM waves				
4	19-20 5(a)	6	derivation	Derive the equations for wave impedance, cut-off frequency, and phase constant for a transverse magnetic wave.
Rectangular waveguide modes				
5	20-21 3(d)	7	numerical	A rectangular waveguide with dimensions $a = 2.5$ cm, $b = 1$ cm is to operate below 15.1 GHz. How many TE and TM modes can the waveguide transmit if the guide is filled with a medium characterized by $\sigma = 0$, $\epsilon = 4\epsilon_0$, $\mu_r = 1$? Calculate the cutoff frequencies of the modes.

4.5. Network analysis:

11 marks, 2 questions, in 1 of 3 papers. Book: Pozar Ch. 4.

Most often theory (1 of 2). Subtopics: S-parameters; Signal flow graphs.

Formulas:

Subtopic	Formula	Equation	Used in
S-parameters (6)	Power waves	$a_n = \dfrac{V_n^+}{\sqrt{Z_0}},\ b_n = \dfrac{V_n^-}{\sqrt{Z_0}}$	19-20 3(a)
	S-parameters	$[b] = [S][a],\ S_{ij} = \left.\dfrac{b_i}{a_j}\right\|_{a_k=0,\, k\neq j}$	19-20 3(a)
	Ideal circulator	$S_{21} = S_{32} = S_{13} = 1$, all other $S_{ij} = 0$	19-20 3(a)
Signal flow graphs (5)	**Input impedance**	$Z_{in} = Z_0 \dfrac{Z_L + jZ_0 \tan\beta l}{Z_0 + jZ_L \tan\beta l}$	19-20 3(c)
	Input reflection	$\Gamma_{in} = \Gamma_L e^{-j2\beta l}$	19-20 3(c)
	Flow graph with the source	$a_1 = \dfrac{b_s}{1 - \Gamma_S \Gamma_L e^{-j2\beta l}}$	19-20 3(c)

Symbols:

a_n, b_n = incident and outgoing waves at port n ($\sqrt{\text{W}}$) ↩

l = line length (m), lossless ↩

b_s = source wave ↩

Γ_S = source reflection coefficient ↩

Questions:

No.	Paper	Marks	Type	Question
S-parameters				
1	19-20 3(a)	6	theory	Why we use S-parameters for microwave analysis? Define S-matrix for 3 port network with appropriate example.
Signal flow graphs				
2	19-20 3(c)	5	derivation	For the figure shown in Fig. 1 draw the flow graph and derive the input impedance formula at port 1 from the signal flow-graph representation.

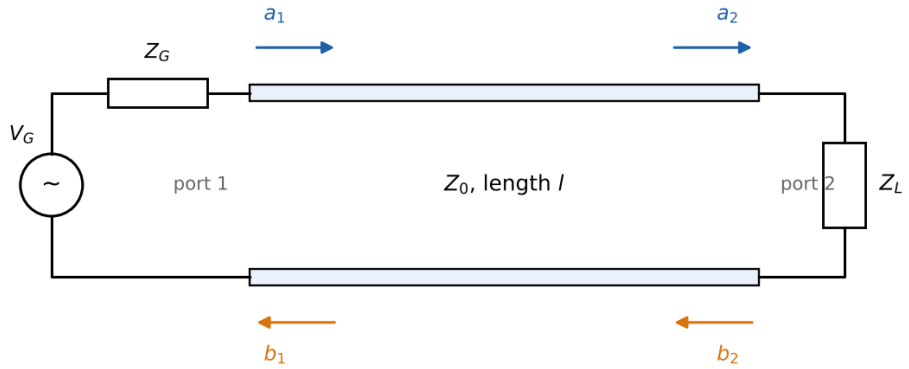


Figure 2: 19-20 3(c): Figure 1 of the paper (redrawn)

4.6. Impedance matching:

43 marks, 7 questions, in 3 of 3 papers. Book: Pozar Ch. 5.

Most often design (4 of 7). Subtopics: Single-stub matching; Quarter-wave transformer.

Formulas:

Subtopic	Formula	Equation	Used in
Single-stub matching (28)	Stub position	$t = \tan\beta d = \dfrac{X_L \pm \sqrt{R_L[(Z_0 - R_L)^2 + X_L^2]/Z_0}}{R_L - Z_0}$	19-20 4(d), 18-19 3(d)
	Short-circuited stub	$\dfrac{l}{\lambda} = \dfrac{1}{2\pi}\tan^{-1}\dfrac{Y_0}{B}$	20-21 2(b), 20-21 3(c), 19-20 4(d), 18-19 3(d)
	Susceptance to cancel	$B = \dfrac{R_L^2 t - (Z_0 - X_L t)(X_L + Z_0 t)}{Z_0\,[R_L^2 + (X_L + Z_0 t)^2]}$	19-20 4(d)
Quarter-wave transformer (15)	**Quarter-wave transformer**	$Z_1 = \sqrt{Z_0 R_L}$	20-21 2(c), 19-20 4(b), 18-19 4(d)

Symbols:

d = distance from the load (m) ↩

R_L, X_L = load resistance and reactance (Ω) ↩

B = susceptance to cancel (S); line susceptance at the stub (S) ↩

l_s = stub length (m)

Z_1 = impedance of the $\lambda/4$ section (Ω) ↩

R_L = real load (Ω); resistive load (Ω) ↩

Questions:

No.	Paper	Marks	Type	Question
Single-stub matching				
1	20-21 2(b)	8	design repeats 18-19 3(d)	A 75 Ω lossless transmission line is connected to a load impedance of $(35 - j50)$ Ω. Find the length position, l_p, and the length, l_s of a short-circuited stub required to match the load. You need to show two possible solutions using Smith Chart.
2	20-21 3(c)	6	design	An antenna with an impednace of $40 + j30$ Ω is to be matched to a 100 Ω lossless line with a shorted stub. Determine: (i) The required stub admittance (ii) The distance between the stub and the antenna (iii) The stub length (iv) The standing wave ratio on each segment of the system
3	19-20 4(d)	6	design repeats 18-19 3(d)	A 50 Ω transmission line is connected to a load impedance of $35 - j47.5$ Ω. Find the position and length of a short-circuited stub required to match the line.
4	18-19 3(d)	8	design	A 50 Ω transmission line is connected to a load impedance $35 - j47.5$ Ω. Find the position and length of a short circuited stub required to match the line.
Quarter-wave transformer				
5	20-21 2(c)	5	theory repeats 18-19 4(d)	Explain how quarter-wave transformer can be used for impdeance matching. Mention its limitations.
6	19-20 4(b)	4	theory repeats 18-19 4(d)	Show how quarter wavelength line is used for matching.
7	18-19 4(d)	6	theory	How is quarter wavelength transmission line used for impedance matching?

4.7. Resonators:

5 marks, 2 questions, in 2 of 3 papers. Book: Pozar Ch. 6.

Most often theory (1 of 2). Subtopics: Cavity resonators.

Formulas:

Subtopic	Formula	Equation	Used in
Cavity resonators (5)	**Cavity resonance**	$f_{mnp} = \frac{c}{2\sqrt{\mu_r \epsilon_r}} \sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2 + \left(\frac{p}{d}\right)^2}$	20-21 6(c), 19-20 4(a)
	Quality factor	$Q = \omega \frac{\text{energy stored}}{\text{power lost}}$	19-20 4(a)

Symbols:

d = cavity length (m) ↩ p = third mode index ↩

Questions:

No.	Paper	Marks	Type	Question
Cavity resonators				
1	20-21 6(c)	3	numerical	A rectangular cavity resonator has dimensions of $a = 5$ cm, $b = 2$ cm, and $d = 15$ cm. Compute the resonant frequency of the dominant mode for a dielectric-field cavity of $\epsilon_r = 2.56$.
2	19-20 4(a)iv	2	theory	Write short notes on the following topic (iv) Microwave cavity resonators

4.8. RF components and filters:

24 marks, 4 questions, in 2 of 3 papers. Book: Ludwig and Bretchko Ch. 1, 4, 5.

Most often derivation (2 of 4). Subtopics: RF behaviour of L and C; Two-port parameters; Filters.

Formulas:

Subtopic	Formula	Equation	Used in
RF behaviour of L and C (12)	High-frequency resistor	$Z = j\omega(2L) + \dfrac{1}{1/R + j\omega C_a}$	18-19 4(a)
	Resistor corners	$f_c = \dfrac{1}{2\pi R C_a},\ f_0 = \dfrac{1}{2\pi\sqrt{2LC_a}}$	18-19 4(a)
	Ideal reactances	$X_L = 2\pi f L,\ X_C = \dfrac{1}{2\pi f C}$	19-20 6(a)
	Capacitor self-resonance	$f_r = \dfrac{1}{2\pi\sqrt{L_s C}}$	19-20 6(a)
	Inductor self-resonance	$f_r = \dfrac{1}{2\pi\sqrt{L C_s}}$	19-20 6(a)
Two-port parameters (6)	ABCD definition	$V_1 = AV_2 + BI_2,\ I_1 = CV_2 + DI_2$	18-19 5(a)
	High-pass ABCD	$A = 1 - \dfrac{1}{\omega^2 LC},\ B = \dfrac{1}{j\omega C},\ C = \dfrac{1}{j\omega L},\ D = 1$	18-19 5(a)
Filters (6)	**High-pass gain**	$H = \dfrac{V_2}{V_1} = \dfrac{j\omega RC}{1 + j\omega RC},\ f_c = \dfrac{1}{2\pi RC}$	19-20 5(b)
	Gain and phase	$\lvert H\rvert = \dfrac{f/f_c}{\sqrt{1 + (f/f_c)^2}},\ \angle H = 90° - \tan^{-1}\dfrac{f}{f_c}$	19-20 5(b)
	Series C in a Z_0 system	$S_{21} = \dfrac{j\omega(2Z_0C)}{1 + j\omega(2Z_0C)}$	19-20 5(b)

Symbols:

L = each lead's inductance (H) ↩ f = frequency (Hz) ↩ C_s = winding capacitance (F) ↩

C_a = end capacitance (F) ↩ L_s = lead inductance (H) ↩ V_1, V_2 = input and output voltages ↩

Questions:

No.	Paper	Marks	Type	Question
RF behaviour of L and C				
1	19-20 6(a)	6	theory	Explain how radio frequency affects the inductor and capacitor reactance.
2	18-19 4(a)	6	diagram	Sketch the equivalent circuit of a high frequency resistor. Also plot a typical frequency versus impedance cure *[sic: should be curve]* of the resistor.
Two-port parameters				
3	18-19 5(a)	6	derivation	For high pass filter determine ABCD parameters of a two port system.
Filters				
4	19-20 5(b)	6	derivation	Find the forward voltage gain of a high-pass filter.

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4.9. Amplifier design:

26 marks, 4 questions, in 2 of 3 papers. Book: Ludwig and Bretchko Ch. 9, 10; Pozar Ch. 12, 13.

Most often theory (3 of 4). Subtopics: Stability; Oscillators; Constant-gain condition; Constant-gain circles.

Formulas:

Subtopic	Formula	Equation	Used in
Stability (5)	Input reflection	$\Gamma_{in} = S_{11} + \dfrac{S_{12}S_{21}\Gamma_L}{1 - S_{22}\Gamma_L}$	18-19 5(b)
	Output stability circle	$C_L = \dfrac{(S_{22} - \Delta S_{11}^*)^*}{\lvert S_{22}\rvert^2 - \lvert\Delta\rvert^2},\ R_L = \left\lvert \dfrac{S_{12}S_{21}}{\lvert S_{22}\rvert^2 - \lvert\Delta\rvert^2} \right\rvert$	18-19 5(b)
	Input stability circle	$C_S = \dfrac{(S_{11} - \Delta S_{22}^*)^*}{\lvert S_{11}\rvert^2 - \lvert\Delta\rvert^2},\ R_S = \left\lvert \dfrac{S_{12}S_{21}}{\lvert S_{11}\rvert^2 - \lvert\Delta\rvert^2} \right\rvert$	18-19 5(b)
Oscillators (7)	Negative-resistance oscillator	$R_{in} + R_L = 0,\ X_{in} + X_L = 0$	18-19 5(c)
	Barkhausen condition	$\lvert A\beta\rvert = 1,\ \angle A\beta = 0°$	18-19 5(c)
Constant-gain condition (5)	Unilateral gain	$G_{TU} = G_S\, G_0\, G_L,\ G_0 = \lvert S_{21}\rvert^2$	19-20 3(b)
	Source gain	$G_S = \dfrac{1 - \lvert\Gamma_S\rvert^2}{\lvert 1 - S_{11}\Gamma_S\rvert^2}$	19-20 3(b)
	Normalized gain	$g_s = \dfrac{G_S}{G_{Smax}} = G_S(1 - \lvert S_{11}\rvert^2)$	19-20 3(b)
	Gain-circle centre	$d_{gs} = \dfrac{g_s S_{11}^*}{1 - \lvert S_{11}\rvert^2(1 - g_s)}$	19-20 3(b)
	Gain-circle radius	$r_{gs} = \dfrac{\sqrt{1 - g_s}\,(1 - \lvert S_{11}\rvert^2)}{1 - \lvert S_{11}\rvert^2(1 - g_s)}$	19-20 3(b)
Constant-gain circles (9)	**Maximum source gain**	$G_{Smax} = \dfrac{1}{1 - \lvert S_{11}\rvert^2}$	19-20 3(d)

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No.	Paper	Marks	Type	Question

Symbols:

S_{ij} = S-parameters

β = feedback ↩

r_{gs} = radius ↩

A = gain ↩

d_{gs} = centre in the Γ_S plane ↩

Questions:

No.	Paper	Marks	Type	Question
Stability				
1	18-19 5(b)	5	theory	What are stability circles? Explain how it is used.
Oscillators				
2	18-19 5(c)	7	theory	Explain the working principle of negative resistance oscillator and feedback oscillator.
Constant-gain condition				
3	19-20 3(b)	5	theory	Explain the condition for constant gain for amplifier.
Constant-gain circles				
4	19-20 3(d)	9	design	A FET is operated at $f = 4$ GHz and is biased such that $s_{11} = 0.7\angle 125°$. It is assumed that the transistor is unconditionally stable. Find the maximum source gain G_{Smax} and sketch the constant gain circles for several values of G_S.

4.10. Microwave tubes and devices:

9 marks, 4 questions, in 1 of 3 papers. Book: Pozar Ch. 11.

Most often theory (4 of 4). Subtopics: Klystron; Magnetron; Gunn diode; Travelling wave tube.

Questions:

No.	Paper	Marks	Type	Question
Klystron				
1	20-21 5(d)i	2.25	theory	Write short notes (any three). (i) Klystron
Magnetron				
2	20-21 5(d)ii	2.25	theory	Write short notes (any three). (ii) Magnetron
Gunn diode				
3	20-21 5(d)iii	2.25	theory	Write short notes (any three). (iii) Gunn Diode Oscillator
Travelling wave tube				
4	20-21 5(d)iv	2.25	theory	Write short notes (any three). (iv) Travelling Wave Tube (TWT)

4.11. Microwave systems:

42 marks, 8 questions, in 3 of 3 papers. Book: Pozar Ch. 14.

Most often theory (3 of 8). Subtopics: Radar; Radio link budget; RF hazards; Receivers and mixers; AM power; EIRP.

Formulas:

Subtopic	Formula	Equation	Used in
Radar (13)	**Radar range**	$R_{max} = \left[\frac{P_t G^2 \lambda^2 \sigma}{(4\pi)^3 P_{min}}\right]^{1/4}$	19-20 4(c), 18-19 2(b)
	Density at the target	$S_t = \frac{P_t G}{4\pi R^2}$	19-20 4(c)
	Radar equation	$P_r = \frac{P_t G^2 \lambda^2 \sigma}{(4\pi)^3 R^4}$	19-20 4(c)

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Subtopic	Formula	Equation	Used in
	Range from delay	$R = \frac{c\,t}{2}$	19-20 4(c)
Radio link budget (11)	**Directivity from aperture**	$D = \frac{4\pi A_e}{\lambda^2}$	18-19 6(c)
	Friis formula	$\frac{P_r}{P_t} = \frac{A_{et}A_{er}}{r^2\lambda^2} = G_tG_r\left(\frac{\lambda}{4\pi r}\right)^2$	18-19 2(c), 18-19 6(c)
	Noise power	$N = kT_{sys}B$	18-19 2(c)
	Link SNR	$\frac{S}{N} = \frac{P_tG_tG_r}{kT_{sys}B}\left(\frac{\lambda}{4\pi r}\right)^2$	18-19 2(c)
RF hazards (2)	Specific absorption rate	$\text{SAR} = \frac{\sigma_t\lvert E\rvert^2}{\rho}$	19-20 4(a)
	Safe distance	$r = \sqrt{\frac{\text{EIRP}}{4\pi S_{lim}}}$	19-20 4(a)
Receivers and mixers (6)	Mixer product	$\cos\omega_{RF}t\cos\omega_{LO}t = \frac{1}{2}\cos(\omega_{RF}-\omega_{LO})t + \frac{1}{2}\cos(\omega_{RF}+\omega_{LO})t$	20-21 5(a)
	Intermediate frequency	$f_{IF} = f_{RF} - f_{LO}$	20-21 5(a)
	Image frequency	$f_{image} = f_{LO} - f_{IF}$	20-21 5(a)
AM power (5)	AM signal	$s(t) = A_c[1+\mu\cos\omega_m t]\cos\omega_c t$	20-21 5(b)
	AM total power	$P_t = P_c\left(1+\frac{\mu^2}{2}\right)$	20-21 5(b)
EIRP (5)	Gain in dB	$G_{dB} = 10\log_{10}G,\ G = 10^{G_{dB}/10}$	20-21 5(c)
	EIRP	$\text{EIRP} = P_tG_t$	20-21 5(c)

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No.	Paper	Marks	Type	Question

Symbols:

σ = radar cross-section (m^2) ↩

P_{min} = minimum detectable signal (W) ↩

P_t = transmitted power (W); transmitter power (W) ↩

G = antenna gain ↩

R = range (m) ↩

t = echo delay (s) ↩

A_e = effective aperture (m^2) ↩

A_{et}, A_{er} = apertures (m^2) ↩

k = Boltzmann's constant (J/K)

T_{sys} = system temperature (K) ↩

B = bandwidth (Hz) ↩

σ_t = tissue conductivity (S/m) ↩

ρ = tissue density (kg/m^3) ↩

S_{lim} = power-density limit (W/m^2) ↩

ω_{RF}, ω_{LO} = signal and local-oscillator frequencies (rad/s) ↩

A_c = carrier amplitude ↩

μ = modulation index ↩

P_c = carrier power (W) ↩

G_{dB} = gain in decibels (dBi) ↩

G_t = transmit antenna gain ↩

Questions:

No.	Paper	Marks	Type	Question
Radar				
1	19-20 4(c)	7	theory repeats 18-19 2(b)	Explain how the RADAR range equation is used to detect object.
2	18-19 2(b)	6	theory	Explain how the RADAR range equation is used to detect objects.
Radio link budget				
3	18-19 2(c)	5	derivation	Show the signal to noise ratio of a radio link is related to bandwidth and system temperature.
4	18-19 6(c)	6	derivation	Show that for antennas with same aperture the received to transmitted power ratio is inversely proportional to square of wavelength and distance between them.
RF hazards				
5	19-20 4(a)iii	2	theory	Write short notes on the following topic (iii) RF radiation hazards and safety standard
Receivers and mixers				
6	20-21 5(a)	6	numerical	An RF signal at $f = 100$ MHz is mixed with local oscillator signal at $f_{LO} = 90$ MHz. (i) Calculate the IF (intermediate frequency) output. (ii) Why are active filters used after mixing in a super-heterodyne receiver?
AM power				
7	20-21 5(b)	5	numerical	For a 1 kHz message signal modulating a 1 MHz carrier with 80% modulation index in AM, calculate the total transmitted power if carrier power is 100 W.
EIRP				
8	20-21 5(c)	5	numerical	For an antenna with gain 10 dB transmitting 10 W power at 2.4 GHz, calculate the effective isotropic radiated power (EIRP) of the antenna.

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4.12. Antennas:

145 marks, 23 questions, in 3 of 3 papers. Book: Kraus; Sadiku Ch. 13.

Most often theory (12 of 23). Subtopics: Antenna aperture; Pattern calculations; Near and far field; Magnetic dipole; Remote sensing; Linear arrays; Reciprocity; Image theory; Polarization; Yagi-Uda antenna; Paraboloid reflectors; Horn antennas; Directivity and beam area; Hertzian dipole; Radiation from accelerating charges; Effective height and area.

Formulas:

Subtopic	Formula	Equation	Used in
Antenna aperture (12)	**Directivity from aperture**	$D = \dfrac{4\pi A_e}{\lambda^2}$	18-19 1(a), 18-19 6(d)
	Effective aperture	$A_e = \dfrac{P_r}{S},\ \varepsilon_{ap} = \dfrac{A_e}{A_p}$	19-20 1(b)
	Aperture and directivity	$D = \dfrac{4\pi A_e}{\lambda^2},\ A_e\,\Omega_A = \lambda^2$	19-20 1(b)
	Aperture beamwidth	$\text{HPBW} \approx \dfrac{51^\circ\,\lambda}{a}$	19-20 1(b)
Pattern calculations (11)	Power and beam area	$P_{rad} = \iint U\,d\Omega,\ \Omega_A = \dfrac{P_{rad}}{U_m}$	18-19 1(b)
	Directivity	$D = \dfrac{4\pi U_m}{P_{rad}} = \dfrac{4\pi}{\Omega_A}$	19-20 1(a), 18-19 1(b)
	Radiated power	$P_{rad} = \displaystyle\int_0^{2\pi}\int_0^{\pi} U \sin\theta\,d\theta\,d\phi$	19-20 1(d)
Near and far field (12)	Field regions	$r_1 = 0.62\sqrt{D^3/\lambda},\ r_2 = 2D^2/\lambda$	19-20 5(d), 18-19 1(c)
Magnetic dipole (8)	Magnetic dipole E	$E_\phi = \dfrac{\eta k^2 IS}{4\pi r}\sin\theta\left[1 + \dfrac{1}{jkr}\right]e^{-jkr}$	18-19 1(d)
	Magnetic dipole H_r	$H_r = \dfrac{jkIS}{2\pi r^2}\cos\theta\left[1 + \dfrac{1}{jkr}\right]e^{-jkr}$	18-19 1(d)
	Magnetic dipole H_θ	$H_\theta = -\dfrac{k^2 IS}{4\pi r}\sin\theta\left[1 + \dfrac{1}{jkr} - \dfrac{1}{(kr)^2}\right]e^{-jkr}$	18-19 1(d)
Remote sensing (10)	Brightness temperature	$T_B = e\,T_{phys}$	18-19 2(a)
	Received noise power	$P = kT_A B$	19-20 1(c), 18-19 2(a)
	Brightness temperature	$T_B = e\,T_{phys},\ e = 1 - \lvert\Gamma\rvert^2$	19-20 1(c)
	Antenna temperature	$T_A = \dfrac{1}{\Omega_A}\iint T_B\,P_n\,d\Omega$	19-20 1(c)
	Radiometer resolution	$\Delta T_{min} = \dfrac{T_{sys}}{\sqrt{B\tau}}$	19-20 1(c)
Linear arrays (16)	Phase difference	$\psi = \beta d\cos\theta + \delta$	19-20 5(c), 18-19 2(d)
	Array factor	$\lvert AF_n\rvert = \left\lvert\dfrac{\sin(N\psi/2)}{N\sin(\psi/2)}\right\rvert$	19-20 5(c), 18-19 2(d)

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Subtopic	Formula	Equation	Used in
	Main beam	$\cos\theta_m = -\dfrac{\delta}{\beta d}$	19-20 5(c), 18-19 2(d)
	Nulls	$\theta_k = \cos^{-1}\left[\dfrac{1}{\beta d}\left(-\delta \pm \dfrac{2k\pi}{N}\right)\right]$	19-20 5(c), 18-19 2(d)
	Broadside beamwidths	$\text{FNBW} \approx \dfrac{2\lambda}{Nd},\ \text{HPBW} \approx 0.886\dfrac{\lambda}{Nd}$	18-19 2(d)
	Broadside beamwidths	$\text{FNBW} = 2\left[\dfrac{\pi}{2} - \cos^{-1}\dfrac{\lambda}{Nd}\right],\ \text{HPBW} \approx 0.886\,\dfrac{\lambda}{Nd}$	19-20 5(c)
	End-fire beamwidth	$\text{FNBW} = 2\cos^{-1}\left(1 - \dfrac{\lambda}{Nd}\right)$	19-20 5(c)
Polarization (3)	Polarization match	$F = \cos^2\gamma$	18-19 6(a)
Paraboloid reflectors (6)	Paraboloid shape	$r = \dfrac{2f}{1+\cos\theta}$	18-19 6(b)
	Paraboloid taper	$\dfrac{S(\theta)}{S(0)} = \left(\dfrac{1+\cos\theta}{2}\right)^2,\ \dfrac{E(\theta)}{E(0)} = \dfrac{1+\cos\theta}{2}$	18-19 6(b)
Horn antennas (7)	**Horn directivity**	$D \approx \dfrac{7.5A_p}{\lambda^2}$	18-19 6(d)
Directivity and beam area (9)	Directivity estimate	$D \approx \dfrac{41\,253}{\theta_{HP}\,\phi_{HP}}$	19-20 1(a)
	Gain	$G = k\,D$	19-20 1(a)
	Beam area	$\Omega_A = \iint P_n(\theta,\phi)\sin\theta\,d\theta\,d\phi \approx \theta_{HP}\,\phi_{HP}$	19-20 1(a)
	Effective height	$h_e = \dfrac{V_{oc}}{E} = \dfrac{1}{I_0}\displaystyle\int I(z)\,dz$	20-21 6(d), 19-20 1(a)
	Height and aperture	$A_e = \dfrac{h_e^2\,Z_0}{4R_r}$	19-20 1(a)
Hertzian dipole (25)	Vector potential	$A_r = \dfrac{\mu_0 Idl}{4\pi r}e^{-j\beta r}\cos\theta,\ A_\theta = -\dfrac{\mu_0 Idl}{4\pi r}e^{-j\beta r}\sin\theta$	20-21 4(a)
	Magnetic field	$\mathbf{H} = \dfrac{Idl}{4\pi}\sin\theta\left[\dfrac{j\beta}{r} + \dfrac{1}{r^2}\right]e^{-j\beta r}\,\hat{a}_\phi$	20-21 4(a)
	Field from **H**	$\mathbf{E} = \dfrac{1}{j\omega\epsilon}\nabla\times\mathbf{H},\ \dfrac{1}{j\omega\epsilon} = \dfrac{\eta}{j\beta}$	20-21 4(b)
	Radial E field	$E_r = \dfrac{\eta Idl}{2\pi}\cos\theta\left[\dfrac{1}{r^2} - \dfrac{j}{\beta r^3}\right]e^{-j\beta r}$	20-21 4(b)
	Polar E field	$E_\theta = \dfrac{\eta Idl}{4\pi}\sin\theta\left[\dfrac{j\beta}{r} + \dfrac{1}{r^2} - \dfrac{j}{\beta r^3}\right]e^{-j\beta r}$	20-21 4(b)
	Average Poynting vector	$\mathbf{P}_{avg} = \dfrac{1}{2}\text{Re}\,(\mathbf{E}\times\mathbf{H}^*)$	20-21 4(c)
	Far-field power density	$\mathbf{P}_{avg} = \dfrac{15\pi I_0^2}{r^2}\left(\dfrac{dl}{\lambda}\right)^2\sin^2\theta\,\hat{a}_r$	20-21 4(c)

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Subtopic	Formula	Equation	Used in
	Radiated power	$P_{rad} = 40\pi^2 \left(\frac{dl}{\lambda}\right)^2 I_0^2$	20-21 4(c)
	Radiation resistance	$R_{rad} = \frac{2P_{rad}}{I_0^2} = 80\pi^2 \left(\frac{dl}{\lambda}\right)^2$	20-21 4(c)
Radiation from accelerating charges (13)	Retarded time	$t_r = t - \frac{R}{c}$	20-21 6(b)
	Retarded potentials	$V = \frac{1}{4\pi\epsilon_0}\int \frac{\rho(t_r)}{R}\,d\tau',\ \mathbf{A} = \frac{\mu_0}{4\pi}\int \frac{\mathbf{J}(t_r)}{R}\,d\tau'$	20-21 6(b)
	Jefimenko, E	$\mathbf{E} = \frac{1}{4\pi\epsilon_0}\int \left[\frac{\rho}{R^2}\hat{\mathbf{R}} + \frac{\dot{\rho}}{cR}\hat{\mathbf{R}} - \frac{\dot{\mathbf{J}}}{c^2 R}\right] d\tau'$	20-21 6(b)
	Jefimenko, B	$\mathbf{B} = \frac{\mu_0}{4\pi}\int \left[\frac{\mathbf{J}}{R^2} + \frac{\dot{\mathbf{J}}}{cR}\right] \times \hat{\mathbf{R}}\,d\tau'$	20-21 6(b)
Effective height and area (4)	**Effective area**	$A_e = \frac{P_{received}}{P_{inc}} = \frac{\lambda^2}{4\pi}G$	20-21 6(d)
	Area from height	$A_e = \frac{\eta\, h_e^2}{4R_{rad}}$	20-21 6(d)

Symbols:

A_e = effective aperture (m^2) ↩

P_r = received power (W) ↩

S = power density (W/m^2); loop area (m^2) ↩

A_p = physical aperture (m^2); mouth area (m^2) ↩

λ = wavelength (m) ↩

a = width of a uniform aperture (m) ↩

U = radiation intensity (W/sr) ↩

P_{rad} = total radiated power (W) ↩

$\sin\theta\, d\theta\, d\phi$ = element of solid angle (sr) ↩

D = largest dimension (m); largest antenna size (m) ↩

e = emissivity ↩

T_A = antenna temperature (K) ↩

k = Boltzmann's constant; radiation efficiency ↩

B = bandwidth (Hz) ↩

T_{phys} = physical temperature (K) ↩

T_{sys} = system noise temperature (K) ↩

τ = integration time (s) ↩

d = element spacing (m) ↩

δ = progressive feed phase (rad) ↩

N = number of elements ↩

2γ = angle on the Poincaré sphere

f = focal length (m) ↩

θ_{HP}, ϕ_{HP} = half-power beamwidths (deg) ↩

P_n = normalized power pattern ↩

V_{oc} = open-circuit voltage (V) ↩

I_0 = terminal current (A) ↩

E = incident field (V/m) ↩

R_r = radiation resistance (Ω) ↩

r, θ = spherical coordinates (m, rad) ↩

$\mathbf{H}$ = magnetic field (A/m) ↩

$\mathbf{P}_{avg}$ = power density (W/m^2) ↩

R_{rad} = radiation resistance (Ω) ↩

R = distance from source point to field point (m) ↩

ρ = charge density (C/m^3) ↩

$\mathbf{J}$ = current density (A/m^2) ↩

P_{inc} = incident power density (W/m^2) ↩

G = antenna gain ↩

Questions:

No.	Paper	Marks	Type	Question
Antenna aperture				
1	19-20 1(b)	6	theory repeats 18-19 1(a)	What is aperture of an Antenna? Show how antenna aperture can be set to increase directivity of an antenna.
2	18-19 1(a)	6	theory	What is aperture of an antenna? Show how antenna aperture can be set to increase directivity of an antenna.
Pattern calculations				
3	19-20 1(d)	5	numerical repeats 18-19 1(b)	Calculate the half power beamwidth, beamwidth between first nulls, radiated power and directivity of an antenna with power pattern $sin\theta sin\varphi$ for $0 \leq \theta \leq \pi,\ 0 \leq \varphi \leq \pi$.
4	18-19 1(b)	6	numerical	Calculate the half power beamwidth, beamwidth between first nulls, radiated power and directivity of an antenna with power pattern $\cos^2\theta \sin\varphi$ for $0 \leq \theta \leq \pi, 0 \leq \varphi \leq \pi$.
Near and far field				
5	19-20 5(d)	7	theory repeats 18-19 1(c)	Define near field and far field for an antenna.
6	18-19 1(c)	5	theory	Describe the field regions surrounding an antenna.
Magnetic dipole				
7	18-19 1(d)	8	derivation	For an elemental magnetic dipole derive the equation for electric field and magnetic field intensity.
Remote sensing				
8	19-20 1(c)	5	theory repeats 18-19 2(a)	Describe how passive remote sensing is used to detect object.
9	18-19 2(a)	5	theory	Describe how passive remote sensing is used to detect object.
Linear arrays				
10	19-20 5(c)	7	derivation repeats 18-19 2(d)	Derive the equation for a uniform linear array and write the conditions for main beam direction, width of main beam and null locations.
11	18-19 2(d)	9	derivation	Derive the equation for a uniform linear array and write the conditions for main beam direction, width of main beam and null locations.
Reciprocity				
12	18-19 3(a)i	3	theory	Write short notes on. (i) Lorentz reciprocity theorem.
Image theory				
13	18-19 3(a)ii	3	theory	Write short notes on. (ii) Image theory.
Polarization				
14	18-19 6(a)i	3	theory	Write short notes on the following topics. (i) Poincare Sphere
Yagi-Uda antenna				
15	18-19 6(a)ii	3	theory	Write short notes on the following topics. (ii) Yagi-Uda antenna.
Paraboloid reflectors				
16	18-19 6(b)	6	derivation	For a paraboloid-of-revolution with an isotropic point source what is the ratio of power density and field intensity ratio.
Horn antennas				
17	18-19 6(d)	7	derivation	For a pyramidal or rectangular horn show that directivity D is $D \approx 10\log\left(\frac{7.5A_p}{\lambda^2}\right)$.
Directivity and beam area				
18	19-20 1(a)	9	theory	Write short notes on the following topics (i) Directivity (ii) Effective Height (iii) Beam Area

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No.	Paper	Marks	Type	Question
Hertzian dipole				
19	20-21 4(a)	8	derivation	The magnetic vector potential of a Hertzian dipole is given by: $\vec{A} = \frac{\mu_0 I dl}{4\pi r} e^{-j\beta r}\hat{z}$. Show that magnetic field, $H = \frac{Idl}{4\pi}\cos(\theta)\left[\frac{j\beta}{r} + \frac{1}{r^2}\right] e^{-j\beta r}\hat{\phi}$ *[sic: should be* $\sin(\theta)$*]*.
20	20-21 4(b)	12	derivation	Using the expression of H found in 4(a), show that the electric field component $E = \frac{\eta I dl}{4\pi}\cos(\theta)\left[\frac{j\beta}{r} + \frac{1}{r^2} - \frac{j}{\beta r^3}\right] e^{-j\beta r}\hat{\theta} + \frac{\eta I dl}{2\pi}\cos(\theta)\left[\frac{1}{r^2} - \frac{j}{\beta r^3}\right] e^{-j\beta r}\hat{r}$ *[sic: the* $\hat{\theta}$ *term should have* $\sin(\theta)$*]*
21	20-21 4(c)	5	derivation	Find the expression of average power density of Hertzian dipole in far field zone.
Radiation from accelerating charges				
22	20-21 6(b)	13	derivation	Derive Jefeminko's equation and from there, explain the necessity of accelerating charge for electromagnetic radiation.
Effective height and area				
23	20-21 6(d)	4	theory repeats 19-20 1(a)	What do you mean by effective height and effective area of an antenna?

5. Theory points:

What a full answer covers, for every theory question, in book order. The link opens the full answer.

No.	Paper	Marks	Subtopic	Points to cover
Transmission lines				
1	19-20 4(a)i	2	Reflection coefficient	$\Gamma = V^-/V^+ = (Z_L - Z_0)/(Z_L + Z_0)$; $\lvert\Gamma\rvert \le 1$ for a passive load; along the line $\Gamma(d) = \Gamma_L e^{-j2\beta d}$; 0 when matched, -1 short, $+1$ open.
2	19-20 4(a)ii	2	Standing wave ratio	$S = \lvert V\rvert_{max}/\lvert V\rvert_{min} = (1+\lvert\Gamma\rvert)/(1-\lvert\Gamma\rvert)$, $1 \le S < \infty$; maxima and minima are $\lambda/4$ apart; $S = 1$ matched, infinite for a short or open.
3	19-20 6(c)	4	Terminated lines	$Z_{sc} = jZ_0 \tan\beta l$, $Z_{oc} = -jZ_0 \cot\beta l$: pure reactances; below $\lambda/4$ a short looks inductive, an open capacitive; they swap every $\lambda/4$ and repeat every $\lambda/2$; lossy: $Z_0 \tanh\gamma l$, $Z_0 \coth\gamma l$; $Z_0 = \sqrt{Z_{oc} Z_{sc}}$.
Smith chart				
4	19-20 2(b)	6	Circle properties	r-circles: centres on the real axis, all through $(1,0)$, shrinking from the unit circle ($r = 0$) to the point $(1,0)$ ($r = \infty$); $r = 1$ through the centre. x-circles: centres on $u = 1$, radius $1/\lvert x\rvert$, all through $(1,0)$; $+x$ above, $-x$ below; the real axis is $x = 0$; the two families cross at right angles.
5	18-19 3(b)	4	Reading impedance	Normalize $z_L = Z_L/Z_0$ and plot it; draw the SWR circle through it; rotate clockwise (towards the generator) by l/λ on the wavelength scale; read z_{in}; multiply by Z_0.
Waveguides and microstrip				
6	20-21 6(a)	5	Microstrip	Uses: RF and microwave PCB interconnects, printed filters, couplers, dividers, matching networks, patch antennas, MICs and MMICs. Advantages: small, light, cheap, planar, easy to mount devices, Z_0 set by the strip width. Losses: conductor (skin effect), dielectric, radiation, surface waves.
7	19-20 2(c)	3	Circular waveguide modes	TE: $E_z = 0$, cutoff from zeros of J_n', dominant TE_{11} ($p'_{11} = 1.841$), $Z_{TE} > \eta$. TM: $H_z = 0$, cutoff from zeros of J_n, lowest TM_{01} ($p_{01} = 2.405$), $Z_{TM} < \eta$.
8	18-19 4(b)	6	Microstrip	Microstrip antenna: a metal patch about half a wavelength long over a ground plane on a dielectric substrate, radiating from its open edges; low profile, cheap, narrow band. Microstrip lines: MIC and PCB interconnects, matching networks, filters, couplers, array feeds.

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No.	Paper	Marks	Subtopic	Points to cover
Network analysis				
9	19-20 3(a)	6	S-parameters	At microwave frequencies opens and shorts are hard to make and can make devices oscillate, while matched loads and wave power are easy to measure (VNA); so networks are described by $b = [S]a$, $S_{ij} = b_i/a_j$ with the other ports matched. Three-port example: a power divider or circulator.
Impedance matching				
10	20-21 2(c)	5	Quarter-wave transformer	A $\lambda/4$ section of impedance Z_1 turns a load R_L into $Z_{in} = Z_1^2/R_L$; with $Z_1 = \sqrt{Z_0 R_L}$, $Z_{in} = Z_0$. Limits: real loads only (or move to a real point first), exact at one frequency (narrow band), Z_1 must be realizable.
11	19-20 4(b)	4	Quarter-wave transformer	A $\lambda/4$ section of impedance Z_1 turns a load R_L into $Z_{in} = Z_1^2/R_L$; with $Z_1 = \sqrt{Z_0 R_L}$, $Z_{in} = Z_0$. Limits: real loads only (or move to a real point first), exact at one frequency (narrow band), Z_1 must be realizable.
12	18-19 4(d)	6	Quarter-wave transformer	A $\lambda/4$ section of impedance Z_1 turns a load R_L into $Z_{in} = Z_1^2/R_L$; with $Z_1 = \sqrt{Z_0 R_L}$, $Z_{in} = Z_0$. Limits: real loads only (or move to a real point first), exact at one frequency (narrow band), Z_1 must be realizable.
Resonators				
13	19-20 4(a)iv	2	Cavity resonators	A closed metal box (or section of guide shorted at both ends) resonates when its length is a whole number of half guide-wavelengths; $f_{mnl} = \frac{c}{2\pi\sqrt{\mu_r \varepsilon_r}}\sqrt{(m\pi/a)^2 + (n\pi/b)^2 + (l\pi/d)^2}$; very high Q; used in filters, oscillators and wavemeters.
RF components and filters				
14	19-20 6(a)	6	RF behaviour of L and C	$X_L = \omega L$ grows and $X_C = 1/\omega C$ falls with frequency, so parasitics matter: a capacitor's lead inductance makes it series-resonant at $1/2\pi\sqrt{L_s C}$ and inductive above; an inductor's winding capacitance makes it parallel-resonant and capacitive above (self-resonant frequency).
15	18-19 4(a)	6	RF behaviour of L and C	R with the parasitic capacitance C_a in parallel and the lead inductance L in series; $\lvert Z\rvert$ is R at low frequency, falls above $f = 1/(2\pi R C_a)$, has a series-resonance minimum, then rises with the lead reactance.
Amplifier design				
16	19-20 3(b)	5	Constant-gain condition	Unilateral amplifier ($S_{12} \approx 0$): $G_{TU} = G_S G_0 G_L$; G_S is constant when Γ_S lies on a circle with centre $d_{gs} = g_s S_{11}^*/[1 - \lvert S_{11}\rvert^2(1 - g_s)]$ and radius $r_{gs} = \sqrt{1 - g_s}(1 - \lvert S_{11}\rvert^2)/[1 - \lvert S_{11}\rvert^2(1 - g_s)]$; same for G_L.
17	18-19 5(b)	5	Stability	Stability circles: the loci on the Γ_L (or Γ_S) plane where $\lvert\Gamma_{in}\rvert = 1$ (or $\lvert\Gamma_{out}\rvert = 1$); one side is stable; choose source and load only in the stable region; unconditional stability when no circle cuts the chart ($K > 1$, $\lvert\Delta\rvert < 1$).
18	18-19 5(c)	7	Oscillators	Negative-resistance oscillator: a device with $R_{in} < 0$ and a resonant load, oscillating when $R_{in} + R_L = 0$ and $X_{in} + X_L = 0$. Feedback oscillator: an amplifier with a frequency-selective feedback network, oscillating when $\lvert A\beta\rvert = 1$ and the loop phase is zero (Barkhausen).
Microwave tubes and devices				
19	20-21 5(d)i	2.25	Klystron	Klystron: electrons velocity-modulated in the buncher cavity bunch in the drift space and give energy to the catcher cavity; high-gain, narrow-band amplifier (reflex klystron as an oscillator).
20	20-21 5(d)ii	2.25	Magnetron	Magnetron: crossed dc E and B fields make electrons rotate as spokes that excite a ring of resonant cavities; efficient high-power oscillator (radar, microwave ovens).

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No.	Paper	Marks	Subtopic	Points to cover
21	20-21 5(d)iii	2.25	Gunn diode	Gunn diode: in n-GaAs above a threshold field electrons move to a heavier valley, giving negative differential mobility; travelling domains make current pulses at a frequency set by the length; low-power microwave source.
22	20-21 5(d)iv	2.25	Travelling wave tube	TWT: an electron beam travels along a helix slow-wave structure at nearly the wave's speed, giving energy to the wave continuously; very wide-band, high-gain amplifier (satellite transponders).
Microwave systems				
23	19-20 4(a)iii	2	RF hazards	RF fields heat tissue (eyes and testes most sensitive); limits set by ICNIRP/IEEE as SAR (W/kg) or power density, stricter for the public than for workers and varying with frequency; keep distance, shield, limit exposure time.
24	19-20 4(c)	7	Radar	Echo power $P_r = P_t G^2 \lambda^2 \sigma / ((4\pi)^3 R^4)$; detection needs $P_r \geq P_{min}$, so $R_{max} = [P_t G^2 \lambda^2 \sigma / ((4\pi)^3 P_{min})]^{1/4}$; range from the echo delay, $R = ct/2$.
25	18-19 2(b)	6	Radar	Echo power $P_r = P_t G^2 \lambda^2 \sigma / ((4\pi)^3 R^4)$; detection needs $P_r \geq P_{min}$, so $R_{max} = [P_t G^2 \lambda^2 \sigma / ((4\pi)^3 P_{min})]^{1/4}$; range from the echo delay, $R = ct/2$.
Antennas				
26	20-21 6(d)	4	Effective height and area	Effective height $h_e = V_{oc}/E$: the length of an equivalent uniform-current antenna (λ/π for a half-wave dipole). Effective area $A_e = P_{received}/S_{incident} = G\lambda^2/4\pi$: the area that collects the received power.
27	19-20 1(a)	9	Directivity and beam area	Directivity $D = U_{max}/U_{av} = 4\pi U_{max}/P_{rad} = 4\pi/\Omega_A$ (isotropic 1, half-wave dipole 1.64). Effective height $h_e = V_{oc}/E$: the length of an equivalent antenna with uniform current equal to the terminal current (λ/π for a half-wave dipole). Beam area $\Omega_A = \iint P_n(\theta, \phi)\, d\Omega$: the solid angle through which all the power would flow at the peak intensity; $D = 4\pi/\Omega_A$; $\Omega_A \approx \theta_{HP}\phi_{HP}$.
28	19-20 1(b)	6	Antenna aperture	Effective aperture $A_e = P_r/S$: the area that collects the received power. $D = 4\pi A_e/\lambda^2 = 4\pi\varepsilon_{ap}A_p/\lambda^2$, so an aperture large in wavelengths and uniformly lit gives a high directivity and a narrow beam.
29	19-20 1(c)	5	Remote sensing	A radiometer's antenna collects the thermal noise objects emit, $T_B = eT_{phys}$; averaged over the beam it gives T_A and power $P = kT_A B$; an object shows as a contrast in T_B against its background (metal looks cold); resolution $\Delta T = T_{sys}/\sqrt{B\tau}$.
30	19-20 5(d)	7	Near and far field	Reactive near field out to $0.62\sqrt{D^3/\lambda}$ ($1/r^2$, $1/r^3$ terms, stored energy); radiating near (Fresnel) field out to $2D^2/\lambda$ (pattern still depends on r); far (Fraunhofer) field beyond, only $1/r$ terms, fixed pattern, $E/H = \eta$.
31	18-19 1(a)	6	Antenna aperture	Effective aperture $A_e = P_r/S$: the area that collects the received power. $D = 4\pi A_e/\lambda^2 = 4\pi\varepsilon_{ap}A_p/\lambda^2$, so an aperture large in wavelengths and uniformly lit gives a high directivity and a narrow beam.
32	18-19 1(c)	5	Near and far field	Reactive near field out to $0.62\sqrt{D^3/\lambda}$ ($1/r^2$, $1/r^3$ terms, stored energy); radiating near (Fresnel) field out to $2D^2/\lambda$ (pattern still depends on r); far (Fraunhofer) field beyond, only $1/r$ terms, fixed pattern, $E/H = \eta$.
33	18-19 2(a)	5	Remote sensing	A radiometer's antenna collects the thermal noise objects emit, $T_B = eT_{phys}$; averaged over the beam it gives T_A and power $P = kT_A B$; an object shows as a contrast in T_B against its background (metal looks cold); resolution $\Delta T = T_{sys}/\sqrt{B\tau}$.
34	18-19 3(a)i	3	Reciprocity	Lorentz reciprocity: $\int E_2 \cdot J_1\, dV = \int E_1 \cdot J_2\, dV$ for linear, isotropic media; so an antenna has the same pattern and impedance transmitting and receiving.

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No.	Paper	Marks	Subtopic	Points to cover
35	18-19 3(a)ii	3	Image theory	Image theory: a perfect conductor plane can be replaced by the mirror image of the sources (horizontal current reversed, vertical current kept) without changing the fields above it.
36	18-19 6(a)i	3	Polarization	Poincaré sphere: each polarization is one point; linear polarizations on the equator, circular at the poles (left and right), elliptical in between; latitude sets the ellipticity, longitude the tilt.
37	18-19 6(a)ii	3	Yagi-Uda antenna	Yagi-Uda: an end-fire array of one driven half-wave dipole, a slightly longer reflector behind it and several shorter directors in front, coupled parasitically; high gain in one direction; TV and VHF/UHF links.

6. Derivations:

Every derivation asked, in book order. Each outline gives the steps; the link opens the full working.

No.	Paper	Marks	Derivation	Outline
Transmission lines				
1	20-21 1(a)	8	Line equations	KVL and KCL on a section Δz; divide by Δz, let it go to 0: telegrapher's equations; phasors; wave equation $V'' = \gamma^2 V$ with $\gamma = \sqrt{(R + j\omega L)(G + j\omega C)}$; low loss: drop RG, $\sqrt{1+x} \approx 1 + x/2$.
2	20-21 1(c)	12	Terminated lines	$Z(z) = V/I$ from the given waves; at $z = 0$, $Z = Z_L$: solve for Γ; put Γ in at $z = l$; group terms; Euler; divide by $2\cos\beta l$.
3	19-20 6(b)	9	Line equations	$I = -(1/(R + j\omega L))\, dV/dz$; differentiate $V_0^{\pm} e^{\mp\gamma z}$; $Z_0 = (R + j\omega L)/\gamma = \sqrt{(R + j\omega L)/(G + j\omega C)}$; at the load $V/I = Z_L$; solve for $\Gamma_L = V_0^-/V_0^+$.
4	18-19 4(c)	7	Terminated lines	General solution $V_0^+(e^{-j\beta z} + \Gamma e^{j\beta z})$; a short gives $\Gamma = -1$; voltage $-2jV_0^+ \sin\beta z$, current $(2V_0^+/Z_0)\cos\beta z$; ratio a distance l from the short: $Z_{in} = jZ_0 \tan\beta l$.
Smith chart				
5	20-21 2(a)	12	Constant-r circles	Write $z = (1+\Gamma)/(1-\Gamma)$ with $\Gamma = u + jv$; rationalize and split real and imaginary parts; complete the square: r-circles centre $(r/(1+r), 0)$, radius $1/(1+r)$; x-circles centre $(1, 1/x)$, radius $1/\lvert x\rvert$.
6	19-20 2(a)	6	Constant-r circles	Write $z = (1+\Gamma)/(1-\Gamma)$ with $\Gamma = u + jv$; rationalize and split real and imaginary parts; complete the square: r-circles centre $(r/(1+r), 0)$, radius $1/(1+r)$; x-circles centre $(1, 1/x)$, radius $1/\lvert x\rvert$.
Waveguides and microstrip				
7	19-20 5(a)	6	TM waves	Maxwell's curl equations give the transverse fields from E_z; wave equation for E_z; separate variables; walls force $\sin(m\pi x/a)\sin(n\pi y/b)$; $\beta = \sqrt{k^2 - k_c^2}$; cutoff at $\beta = 0$; transverse fields; $Z_{TM} = \eta\beta/k$.
Network analysis				
8	19-20 3(c)	5	Signal flow graphs	One equation per branch; draw the flow graph; Γ_{in} = path from a_1 to b_1: $\Gamma_L e^{-j2\beta l}$; $Z_{in} = Z_0(1+\Gamma_{in})/(1-\Gamma_{in})$; put in Γ_L, Euler, divide by $2\cos\beta l$.
RF components and filters				
9	19-20 5(b)	6	Filters	Voltage division $R/(R + 1/j\omega C)$; write as $j(f/f_c)/(1 + jf/f_c)$; magnitude and phase; $G_V = 20\log\lvert H\rvert$.
10	18-19 5(a)	6	Two-port parameters	ABCD of a series impedance $[1, Z; 0, 1]$ and of a shunt admittance $[1, 0; Y, 1]$; cascade by matrix product; put in $Z = 1/(j\omega C)$ and $Y = 1/(j\omega L)$.
Microwave systems				
11	18-19 2(c)	5	Radio link budget	Power density $P_t G_t/(4\pi r^2)$; received power with $A_e = G_r\lambda^2/4\pi$; noise power $kT_{sys}B$; divide: $S/N = P_t G_t G_r \lambda^2/((4\pi r)^2 kT_{sys}B)$.

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No.	Paper	Marks	Derivation	Outline
12	18-19 6(c)	6	Radio link budget	Isotropic density $P_t/(4\pi r^2)$; directive antenna multiplies by $D = 4\pi A/\lambda^2$; received power = density $\times$ A; ratio $P_r/P_t = A^2/(r^2\lambda^2)$ (Friis for equal apertures).
Antennas				
13	20-21 4(a)	8	Hertzian dipole	Write $\mathbf{A}$ in spherical components; curl in spherical coordinates (only the ϕ part survives); the two derivatives; combine; divide by μ_0.
14	20-21 4(b)	12	Hertzian dipole	$\mathbf{E} = (1/j\omega\varepsilon)\nabla \times \mathbf{H}$ with only H_ϕ; radial part from $\partial(\sin\theta H_\phi)/\partial\theta$; polar part from $\partial(rH_\phi)/\partial r$; write with η and β.
15	20-21 4(c)	5	Hertzian dipole	Far field keeps the $1/r$ terms: $E_\theta = \eta H_\phi$; $\mathbf{P}_{avg} = \frac{1}{2}\mathrm{Re}(\mathbf{E} \times \mathbf{H}^*)$; put in $\lvert H_\phi\rvert$; free space $\eta = 120\pi$.
16	20-21 6(b)	13	Radiation from accelerating charges	Retarded potentials; $\nabla t_r = -\hat{R}/c$ and $\nabla(1/R) = -\hat{R}/R^2$; gradient of V and time derivative of $\mathbf{A}$ give $\mathbf{E}$; curl of $\mathbf{A}$ gives $\mathbf{B}$; the $1/R$ terms need $\dot{\rho}$ and $\dot{\mathbf{J}}$: radiation needs accelerating charges.
17	19-20 5(c)	7	Linear arrays	Phase step $\psi = \beta d\cos\theta + \delta$; add N elements: geometric series; sum it; write with sines; normalize: $\lvert AF_n\rvert = \lvert\sin(N\psi/2)/(N\sin(\psi/2))\rvert$; main beam at $\psi = 0$, nulls at $\psi = \pm 2k\pi/N$.
18	18-19 1(d)	8	Magnetic dipole	Vector potential of a small current loop; $H = \nabla \times A/\mu$ in spherical coordinates; E from $\nabla \times H = j\omega\varepsilon E$; collect the terms in $1/r$, $1/r^2$, $1/r^3$; far field keeps $1/r$.
19	18-19 2(d)	9	Linear arrays	Phase step $\psi = \beta d\cos\theta + \delta$; add N elements: geometric series; sum it; write with sines; normalize: $\lvert AF_n\rvert = \lvert\sin(N\psi/2)/(N\sin(\psi/2))\rvert$; main beam at $\psi = 0$, nulls at $\psi = \pm 2k\pi/N$.
20	18-19 6(b)	6	Paraboloid reflectors	Parabola $r = 2f/(1+\cos\theta)$; radius on the aperture $\rho = r\sin\theta$; power in a cone maps to an annulus; ratio to the centre $S(\theta)/S(0) = [(1+\cos\theta)/2]^2$; field ratio its square root.
21	18-19 6(d)	7	Horn antennas	$D = 4\pi A_e/\lambda^2$; $A_e = \varepsilon_{ap}A_p$; a horn has $\varepsilon_{ap} \approx 0.6$; $4\pi \times 0.6 \approx 7.5$; in dB $10\log(7.5A_p/\lambda^2)$.

7. Design recipes:

Every design asked, in book order. Each recipe gives the steps in the order to take them; the link opens the full working.

No.	Paper	Marks	Design	Recipe
Impedance matching				
1	20-21 2(b)	8	Single-stub matching	Normalize the load and convert to admittance; move along the SWR circle towards the generator to the $g = 1$ circle (two points); the distance is d; the stub cancels the susceptance; a short-circuited stub of that susceptance gives its length.
2	20-21 3(c)	6	Single-stub matching	Normalize the load and convert to admittance; move along the SWR circle towards the generator to the $g = 1$ circle (two points); the distance is d; the stub cancels the susceptance; a short-circuited stub of that susceptance gives its length.
3	19-20 4(d)	6	Single-stub matching	Normalize the load and convert to admittance; move along the SWR circle towards the generator to the $g = 1$ circle (two points); the distance is d; the stub cancels the susceptance; a short-circuited stub of that susceptance gives its length.
4	18-19 3(d)	8	Single-stub matching	Normalize the load and convert to admittance; move along the SWR circle towards the generator to the $g = 1$ circle (two points); the distance is d; the stub cancels the susceptance; a short-circuited stub of that susceptance gives its length.
Amplifier design				
5	19-20 3(d)	9	Constant-gain circles	$G_{Smax} = 1/(1-\lvert S_{11}\rvert^2)$ at $\Gamma_S = S_{11}^*$; for each chosen G_S the normalized $g_s = G_S/G_{Smax}$; circle centre d_{gs} and radius r_{gs}; draw them on the line at the angle of S_{11}^*.

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8. Numericals:

Every numerical asked, in book order: the method in one line and the final answer, so a worked attempt can be checked. The link opens the full working.

No.	Paper	Marks	Subtopic	Method	Answer
EM waves					
1	20-21 3(a)	6	Plane waves	Read ω and β from the argument $\omega t+\beta z$ (+ sign: travels in $-z$); $\lambda=2\pi/\beta$, $f=\omega/2\pi$; $\varepsilon_r=(\beta c/\omega)^2$; $\eta=120\pi/\sqrt{\varepsilon_r}$; $\vec{H}$ with $\vec{E}\times\vec{H}$ along the propagation.	$-\hat{a}_z$; $\lambda=3.142$ m, $f=15.92$ MHz, $\varepsilon_r=36$; $\vec{H}=0.796\sin(10^8t+2z)\hat{a}_x$ A/m
Transmission lines					
2	20-21 1(b)	5	Line constants	γ into α, β; ω from $\beta=\omega\sqrt{LC}$; $Z_0=\sqrt{L/C}$; $\alpha_c=R/2Z_0$; $G=2(\alpha-\alpha_c)/Z_0$; check $R\ll\omega L$, $G\ll\omega C$.	$G=0.02004$ S/m (about 796 MHz)
3	20-21 3(b)	6	Line constants	$Z_0=\sqrt{(R+j\omega L)/(G+j\omega C)}$; $\gamma=\sqrt{(R+j\omega L)(G+j\omega C)}$; $u_p=\omega/\beta$.	$Z_0=70.75\angle-1.37^\circ\ \Omega$; $\gamma=0.212+j8.888$ /km; $u_p=7.07\times10^5$ m/s
4	19-20 2(d)	10	Lossy lines	$Z_{in}=Z_0\tanh\gamma l$ for a short: solve $\tanh\gamma l=Z_{in}/Z_0$ for γl (inverse tanh); α and β from its parts; then the general formula with the new load.	$\alpha=0.0279$ Np/m, $\beta=0.509$ rad/m; $Z_{in}=185.9-j28.8\ \Omega$
5	19-20 6(d)	7	Slotted-line measurement	$\lambda=2\times$ minima spacing; $\lvert\Gamma\rvert=(S-1)/(S+1)$; phase from the first minimum, $\theta_\Gamma=2\beta d_{min}-\pi$; $Z_L=Z_0(1+\Gamma)/(1-\Gamma)$.	$\Gamma_L=0.333\angle-60^\circ$; $Z_L=57.1-j37.1\ \Omega$
6	18-19 3(c)	7	Terminated lines	$\Gamma_L=(Z_L-Z_0)/(Z_L+Z_0)$; SWR $=(1+\lvert\Gamma\rvert)/(1-\lvert\Gamma\rvert)$; first voltage maximum where the phase of $\Gamma(d)=\Gamma_L e^{-j2\beta d}$ is 0: $d=\theta_\Gamma\lambda/(4\pi)$.	$\Gamma_L=0.598\angle21.8^\circ$; SWR $=3.98$; first maximum 0.0303λ from the load
7	18-19 5(d)	7	Open and short tests	$Z_0=\sqrt{Z_{oc}Z_{sc}}$; $\tan\beta l=\sqrt{Z_{sc}/Z_{oc}}$ (imaginary parts); for the doubled length use $Z_{sc}=jZ_0\tan2\beta l$, $Z_{oc}=-jZ_0\cot2\beta l$.	$Z_0=75\ \Omega$; $\gamma=j0.628$ m^{-1}; doubled: $-j232\ \Omega$ (short), $+j24.2\ \Omega$ (open)
Waveguides and microstrip					
8	20-21 3(d)	7	Rectangular waveguide modes	$u'=1/\sqrt{\mu\varepsilon}$; $f_c=(u'/2)\sqrt{(m/a)^2+(n/b)^2}$; list TE ($m$ or n may be 0) and TM ($m,n\geq1$) modes below 15.1 GHz.	11 TE and 4 TM modes; TE_{10} at 3 GHz
Resonators					
9	20-21 6(c)	3	Cavity resonators	$u'=c/\sqrt{\varepsilon_r}$; dominant mode TE_{101} since $d>a>b$; $f_r=(u'/2)\sqrt{(1/a)^2+(1/d)^2}$.	1.976 GHz
Microwave systems					
10	20-21 5(a)	6	Receivers and mixers	$f_{IF}=f_{RF}-f_{LO}$; the sum and the image ($f_{LO}-f_{IF}$) are rejected by filters.	$f_{IF}=10$ MHz; filters pick the IF and reject the sum, leakage and adjacent channels
11	20-21 5(b)	5	AM power	$P_t=P_c(1+\mu^2/2)$; each sideband $P_c\mu^2/4$.	132 W (16 W per sideband)
12	20-21 5(c)	5	EIRP	Gain from dB, $G=10^{G_{dB}/10}$; EIRP $=P_tG$.	100 W = 20 dBW = 50 dBm

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No.	Paper	Marks	Subtopic	Method	Answer
Antennas					
13	19-20 1(d)	5	Pattern calculations	Find where $U = U_m/2$ and $U = 0$ in the principal planes; $P_{rad} = \iint U \sin\theta \, d\theta \, d\phi$; $D = 4\pi U_m / P_{rad}$.	HPBW = 120°, FNBW = 180°; $P_{rad} = \pi$ W; $D = 4$ (6.02 dBi)
14	18-19 1(b)	6	Pattern calculations	Find where $U = U_m/2$ (HPBW) and $U = 0$ (FNBW) in the principal plane; $P_{rad} = \iint U \sin\theta \, d\theta \, d\phi$ over the given ranges; $D = 4\pi U_m / P_{rad}$.	HPBW = 90°, FNBW = 180°; $P_{rad} = 4/3$ W; $D = 9.42$ (9.74 dBi)

9. Diagrams to practise:

Every figure an answer needs, whatever the question's type, in book order. Draw each one from memory, then check it against the solution.

No.	Paper	Kind	Diagram
EM waves			
1	20-21 3(a)	Diagram	the plane wave with E, H and propagation
Transmission lines			
2	20-21 1(a)	Circuit	lumped-element model of a short line
3	20-21 1(a)	Waveform	travelling wave without and with loss
4	20-21 1(c)	Diagram	line with load and Z_{in} (Fig. 1)
5	20-21 1(c)	Graph	standing-wave patterns
6	19-20 4(a)ii	Graph	standing-wave patterns for three loads
7	19-20 6(b)	Circuit	lumped model of a short piece of line
8	19-20 6(b)	Diagram	line with the load at $z = 0$
9	19-20 6(c)	Graph	input reactance of short- and open-circuited lines
10	19-20 6(d)	Graph	measured standing wave and the equivalent line
11	18-19 3(c)	Diagram	the load on a Smith chart
12	18-19 4(c)	Graph	voltage, current and reactance along a short-circuited line
Smith chart			
13	20-21 2(a)	Diagram	z plane mapped into the unit circle
14	20-21 2(a)	Diagram	constant-r circles and constant-x arcs
15	19-20 2(a)	Diagram	z plane mapped into the unit circle
16	19-20 2(a)	Diagram	constant-r circles and constant-x arcs
17	19-20 2(b)	Diagram	the landmarks of the Smith chart
Waveguides and microstrip			
18	20-21 3(d)	Diagram	TE10 field across the guide
19	20-21 3(d)	Graph	cutoff frequencies of the filled guide
20	20-21 6(a)	Diagram	microstrip cross-section and field
21	19-20 2(c)	Diagram	TE11 and TM01 fields in a circular guide
22	19-20 5(a)	Diagram	rectangular guide
23	19-20 5(a)	Graph	TM β and impedance against frequency
24	18-19 4(b)	Diagram	microstrip patch fed by a line
Network analysis			
25	19-20 3(a)	Diagram	three-port network with incident and outgoing waves
26	19-20 3(c)	Diagram	signal flow graph of the loaded line
Impedance matching			
27	20-21 2(b)	Circuit	single shunt stub
28	20-21 2(b)	Diagram	Smith chart solution
29	20-21 2(c)	Diagram	quarter-wave transformer between line and load

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No.	Paper	Kind	Diagram
30	20-21 2(c)	Graph	reflection against frequency (bandwidth)
31	20-21 3(c)	Circuit	stub matching of the antenna
32	20-21 3(c)	Diagram	Smith chart solution
33	19-20 4(b)	Diagram	quarter-wave transformer between line and load
34	19-20 4(b)	Graph	reflection against frequency (bandwidth)
35	19-20 4(d)	Circuit	stub between load and generator
36	19-20 4(d)	Diagram	single-stub matching on the Smith chart
37	18-19 3(d)	Circuit	stub between load and generator
38	18-19 3(d)	Diagram	single-stub matching on the Smith chart
39	18-19 4(d)	Diagram	quarter-wave transformer between line and load
40	18-19 4(d)	Graph	reflection against frequency (bandwidth)
Resonators			
41	20-21 6(c)	Diagram	cavity and its dominant mode
42	19-20 4(a)iv	Diagram	rectangular cavity and its TE101 field
RF components and filters			
43	19-20 5(b)	Circuit	RC high-pass filter
44	19-20 5(b)	Graph	its gain
45	19-20 6(a)	Circuit	real L and C models at RF
46	18-19 4(a)	Circuit	high-frequency resistor model
47	18-19 4(a)	Graph	impedance of a resistor against frequency
48	18-19 5(a)	Circuit	LC high-pass section
Amplifier design			
49	19-20 3(b)	Block diagram	unilateral amplifier with input and output matching networks
50	19-20 3(d)	Diagram	constant source-gain circles on the Smith chart
51	18-19 5(b)	Diagram	output stability circle on the Γ_L plane
52	18-19 5(c)	Block diagram	negative-resistance and feedback oscillators
Microwave tubes and devices			
53	20-21 5(d)i	Diagram	two-cavity klystron
54	20-21 5(d)i	Graph	Applegate diagram
55	20-21 5(d)ii	Diagram	cavity magnetron cross-section
56	20-21 5(d)iii	Graph	Gunn effect velocity-field curve
57	20-21 5(d)iv	Diagram	helix travelling-wave tube
Microwave systems			
58	20-21 5(a)	Block diagram	superheterodyne receiver
59	20-21 5(a)	Spectrum	frequencies around the mixer
60	20-21 5(b)	Waveform	AM wave and its power spectrum
61	20-21 5(c)	Diagram	directive antenna and its isotropic equivalent
62	19-20 4(a)iii	Graph	ICNIRP reference levels for power density
63	19-20 4(c)	Diagram	radar geometry, outgoing wave and echo
64	18-19 2(b)	Diagram	radar geometry, outgoing wave and echo
65	18-19 2(c)	Diagram	radio link with signal and noise at the receiver
Antennas			
66	20-21 4(a)	Diagram	Hertzian dipole and spherical coordinates
67	20-21 4(b)	Graph	how the $1/r$, $1/r^2$ and $1/r^3$ terms compare with distance
68	20-21 4(c)	Graph	radiation pattern of a Hertzian dipole
69	20-21 6(b)	Diagram	source point, field point and retarded time
70	20-21 6(b)	Diagram	why a stopping charge radiates
71	20-21 6(d)	Diagram	effective height of a half-wave dipole
72	19-20 1(a)	Diagram	real pattern and its equivalent beam-area cone

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No.	Paper	Kind	Diagram
73	19-20 1(a)	Diagram	effective height of a half-wave dipole
74	19-20 1(b)	Diagram	bigger aperture, narrower beam
75	19-20 1(c)	Diagram	passive remote sensing with a radiometer
76	19-20 1(d)	Graph	the pattern in its two principal planes
77	19-20 5(c)	Diagram	array geometry and array factor
78	19-20 5(d)	Graph	the three field terms against distance
79	19-20 5(d)	Diagram	field regions around an antenna
80	18-19 1(a)	Diagram	bigger aperture, narrower beam
81	18-19 1(b)	Graph	the pattern in two cuts
82	18-19 1(c)	Graph	the three field terms against distance
83	18-19 1(c)	Diagram	field regions around an antenna
84	18-19 1(d)	Diagram	small current loop and the field point
85	18-19 2(a)	Diagram	passive remote sensing with a radiometer
86	18-19 2(d)	Diagram	array geometry and array factor
87	18-19 3(a)ii	Diagram	sources above a conductor and their images
88	18-19 6(a)i	Diagram	the Poincaré sphere
89	18-19 6(a)ii	Diagram	Yagi-Uda antenna
90	18-19 6(b)	Diagram	paraboloid with the source at the focus
91	18-19 6(d)	Diagram	pyramidal horn

10. Comparisons:

Every comparison asked, in book order, with the key difference in one line. A pair asked in several papers is one row.

Pair	Key difference	Asked in
Transmission lines		
Short-circuited vs open-circuited line	$jZ_0 \tan \beta l$, inductive below $\lambda/4$ vs $-jZ_0 \cot \beta l$, capacitive below $\lambda/4$	19-20 6(c)
Smith chart		
r-circles vs x-circles	centres on the real axis, inside the chart vs centres on $u = 1$, arcs above and below the axis	19-20 2(b)
Waveguides and microstrip		
TE vs TM modes	$E_z = 0$, cutoff from zeros of J_n', $Z > \eta$ vs $H_z = 0$, cutoff from zeros of J_n, $Z < \eta$	19-20 2(c)
Network analysis		
S-parameters vs Z/Y parameters	waves with matched ports, measurable at microwave vs voltages and currents with open or short ports	19-20 3(a)
Amplifier design		
Negative-resistance vs feedback oscillator	one-port device with $R_{in} < 0$ vs amplifier with a feedback loop, $\lvert A\beta \rvert = 1$	18-19 5(c)
Microwave tubes and devices		
Klystron vs TWT	resonant cavities, narrow band vs helix slow-wave line, very wide band	20-21 5(d)iv
Antennas		
Near field vs far field	reactive $1/r^2$, $1/r^3$ terms, pattern changes with r vs only $1/r$ terms, fixed pattern, $E/H = \eta$	19-20 5(d), 18-19 1(c)
Passive vs active remote sensing	measures natural thermal emission vs sends a signal and measures the echo (radar)	19-20 1(c), 18-19 2(a)
Effective height vs effective area	open-circuit voltage per field (m) vs received power per power density (m^2)	20-21 6(d)

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11. Repeat tracker:

Questions that came back, exactly or with new numbers.

Paper	Asked before as	Topic	Subtopic
2020-21 2(a)	2019-20 2(a)	Smith chart	Constant-r circles
2020-21 2(b)	2018-19 3(d)	Impedance matching	Single-stub matching
2020-21 2(c)	2018-19 4(d)	Impedance matching	Quarter-wave transformer
2020-21 6(a)	2018-19 4(b)	Waveguides and microstrip	Microstrip
2020-21 6(d)	2019-20 1(a)	Antennas	Effective height and area
2019-20 1(b)	2018-19 1(a)	Antennas	Antenna aperture
2019-20 1(c)	2018-19 2(a)	Antennas	Remote sensing
2019-20 1(d)	2018-19 1(b)	Antennas	Pattern calculations
2019-20 4(b)	2018-19 4(d)	Impedance matching	Quarter-wave transformer
2019-20 4(c)	2018-19 2(b)	Microwave systems	Radar
2019-20 4(d)	2018-19 3(d)	Impedance matching	Single-stub matching
2019-20 5(c)	2018-19 2(d)	Antennas	Linear arrays
2019-20 5(d)	2018-19 1(c)	Antennas	Near and far field

12. Question style:

How each topic was asked, as a number of questions.

Topic	Theory	Derivation	Numerical	Diagram	Design
EM waves	·	·	1	·	·
Transmission lines	3	4	6	·	·
Smith chart	2	2	·	·	·
Waveguides and microstrip	3	1	1	·	·
Network analysis	1	1	·	·	·
Impedance matching	3	·	·	·	4
Resonators	1	·	1	·	·
RF components and filters	1	2	·	1	·
Amplifier design	3	·	·	·	1
Microwave tubes and devices	4	·	·	·	·
Microwave systems	3	2	3	·	·
Antennas	12	9	2	·	·

13. Study order:

Topics by marks offered, with the share of all marks covered so far. Papers offer more marks than you answer, so this is a guide to where marks have been, not a score.

No.	Topic	Marks	Covered so far
1	Antennas	145	32 %
2	Transmission lines	86	51 %
3	Impedance matching	43	61 %
4	Microwave systems	42	70 %
5	Smith chart	28	76 %
6	Waveguides and microstrip	27	82 %
7	Amplifier design	26	88 %
8	RF components and filters	24	93 %
9	Network analysis	11	96 %
10	Microwave tubes and devices	9	98 %

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No.	Topic	Marks	Covered so far
11	EM waves	6	99 %
12	Resonators	5	100 %

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